

Belgian Chocolate Problem

A Proposed Exact Algorithmic Solution

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Proof candidate v1.0.0 — 13 September 2026

The problem. Determine all $\delta > 0$ for which nonzero real polynomials x, y, p exist with every root strictly in the left half-plane, $\deg y \leq \deg x$, and

$$p(s) = (s^2 - 2\delta s + 1)x(s) + (s^2 - 1)y(s).$$

No bound on degree or special factorization is imposed. Nonzero constants are permitted. This is the polynomial simultaneous-stabilization formulation.

Main result. Set $q = (1 - \delta)/(1 + \delta)$. The candidate identifies a real $Q \in (0, 1/2)$ through a certified, terminating arbitrary-precision algorithm: for each natural m , it returns rational L, R with

$$L < Q < R, \quad R - L < 2^{-m}.$$

Writing $\Delta = (1 - Q)/(1 + Q)$, the original feasible set is exactly $\mathcal{A} = (0, \Delta)$, and Δ is infeasible. For $0 < q < 1$, analytic feasibility is $Q \leq q$, whereas original polynomial feasibility is $Q < q$. The analytic endpoint is attained; the polynomial endpoint is independently excluded.

Core mechanism. The analytic class consists of real-symmetric holomorphic functions on the unit disc with exactly the simple zero 0 and the two simple 1-points $\pm i\sqrt{q}$. Explicit uniform four-factor coefficient bounds yield

$$\text{analytic feasibility} \iff \text{every finite semialgebraic level } H_N(q) \text{ is feasible} \quad (0 < q \leq 9/16).$$

Compactness supplies compatible infinite coefficients and exact analytic identities. Finite negative obstructions and finite rational positive certificates give searches on both strict sides. Two distinct trisection points prevent endpoint equality from blocking termination. Strict-slack polynomial realization and an actual-degree-preserving perturbation complete the original interface.

Formal verification. Lean 4.33.0 / Mathlib 4.33.0. The concrete main theorem is

`BelgianChocolate.belgianChocolate_exact_algorithmic_solution`

in `BCPThreshold/FinalMain.lean`. The main theorem has no pending mathematical-interface hypothesis. The build and audit pass with zero proof holes and no project-defined axioms; its transitive axioms are `propext`, `Classical.choice`, and `Quot.sound`. Run `lake exe cache get` and `./verify_bcp.sh` from the release. This verifies the main conclusion and its dependency graph, not every paper assertion verbatim. Finite scans are executable; the mathematical iterator uses noncomputable `Nat.find`.

Scope and review request. No closed form, algebraicity, minimal polynomial, special-function formula, practical complexity bound, fixed- N decision, or black-box equality test $q = Q$ is claimed. No native threshold calculator is supplied. AI models assisted exploration, adversarial review and Lean engineering. Independent verification, especially of the original formulation and the analytic/polynomial interface, is actively requested.

Links. Paper GitHub Release Contact / independent review

<https://github.com/chengxuandi/belgian-chocolate-problem-lean-proof-20260910210030-init>